DARSI VAMSI
COMET FWC 071

Implementation of Three LED Blinking using Direct GPIO Register Programming on the VAMAN ARM Development Board

1 Objective

The objective of this experiment is to understand the operation of General Purpose Input/Output (GPIO) registers by directly programming the GPIO peripheral of the VAMAN ARM Development Board. The experiment demonstrates sequential blinking of three LEDs using memory-mapped register access without relying on external hardware abstraction libraries.

2 Hardware Requirements

1. VAMAN ARM Development Board
2. USB Programming Cable
3. Ubuntu 24.04 LTS
4. Three On-board LEDs

3 Software Requirements

1. GNU ARM Embedded GCC Compiler
2. Make Utility
3. Makefile
4. Text Editor (VS Code / Vim)

5. Terminal

4 Theory

General Purpose Input Output (GPIO) is one of the most fundamental peripherals available in ARM-based microcontrollers. GPIO pins can be configured either as input pins for reading external signals or as output pins for driving external devices such as LEDs.

In this experiment, GPIO registers are accessed directly through memory-mapped addresses. The GPIO Direction Register configures the direction of each GPIO pin, while the GPIO Output Register controls the logic level present on the configured output pins.

The experiment avoids the use of any hardware abstraction library and directly manipulates the peripheral registers using pointers.

5 GPIO Register Mapping

Register	Address Offset	Purpose
GPIO_DIR	0x00	Configure GPIO Direction
GPIO_OUT	0x04	GPIO Output Register

Table 1: GPIO Register Description

6 LED Connections

7 Program Description

The application begins by defining the base address of the GPIO peripheral and creating register definitions using volatile pointers.

LED	GPIO Pin
LED 1	GPIO18
LED 2	GPIO21
LED 3	GPIO22

Table 2: GPIO Pins Used

Three GPIO pins (18, 21 and 22) are selected for controlling the LEDs.

Initially, the GPIO direction register is configured so that all three pins operate as output pins.

The program continuously performs the following operations:

1. Turn ON LED 1.
2. Wait using a software delay.
3. Turn OFF LED 1.
4. Turn ON LED 2.
5. Wait.
6. Turn OFF LED 2.
7. Turn ON LED 3.
8. Wait.
9. Turn OFF LED 3.
10. Repeat indefinitely.

8 Algorithm

1. Define GPIO register addresses.
2. Configure GPIO18, GPIO21 and GPIO22 as output pins.
3. Enter an infinite loop.

4. Turn ON LED1.
5. Execute software delay.
6. Turn OFF LED1.
7. Turn ON LED2.
8. Execute software delay.
9. Turn OFF LED2.
10. Turn ON LED3.
11. Execute software delay.
12. Turn OFF LED3.
13. Repeat forever.

9 Source Code

10 Testing Procedure

1. Compile the source code using the ARM GCC compiler.
2. Generate the executable binary.
3. Flash the binary into the VAMAN ARM Development Board.
4. Reset the development board.
5. Observe the three LEDs.
6. Verify that each LED glows one after another with a fixed delay.

11 Expected Output

The LEDs blink in the following order:

$$\boxed{\text{LED1} \rightarrow \text{LED2} \rightarrow \text{LED3} \rightarrow \text{Repeat}}$$

Each LED remains ON for approximately the duration of the software delay before switching to the next LED.

12 Observation

- GPIO registers were successfully configured using direct memory-mapped register programming.
- GPIO18, GPIO21 and GPIO22 functioned correctly as digital output pins.
- Each LED blinked sequentially according to the program logic.
- The software delay generated a visible blinking interval.
- The experiment verified successful low-level GPIO programming on the VAMAN ARM Development Board.

13 Advantages

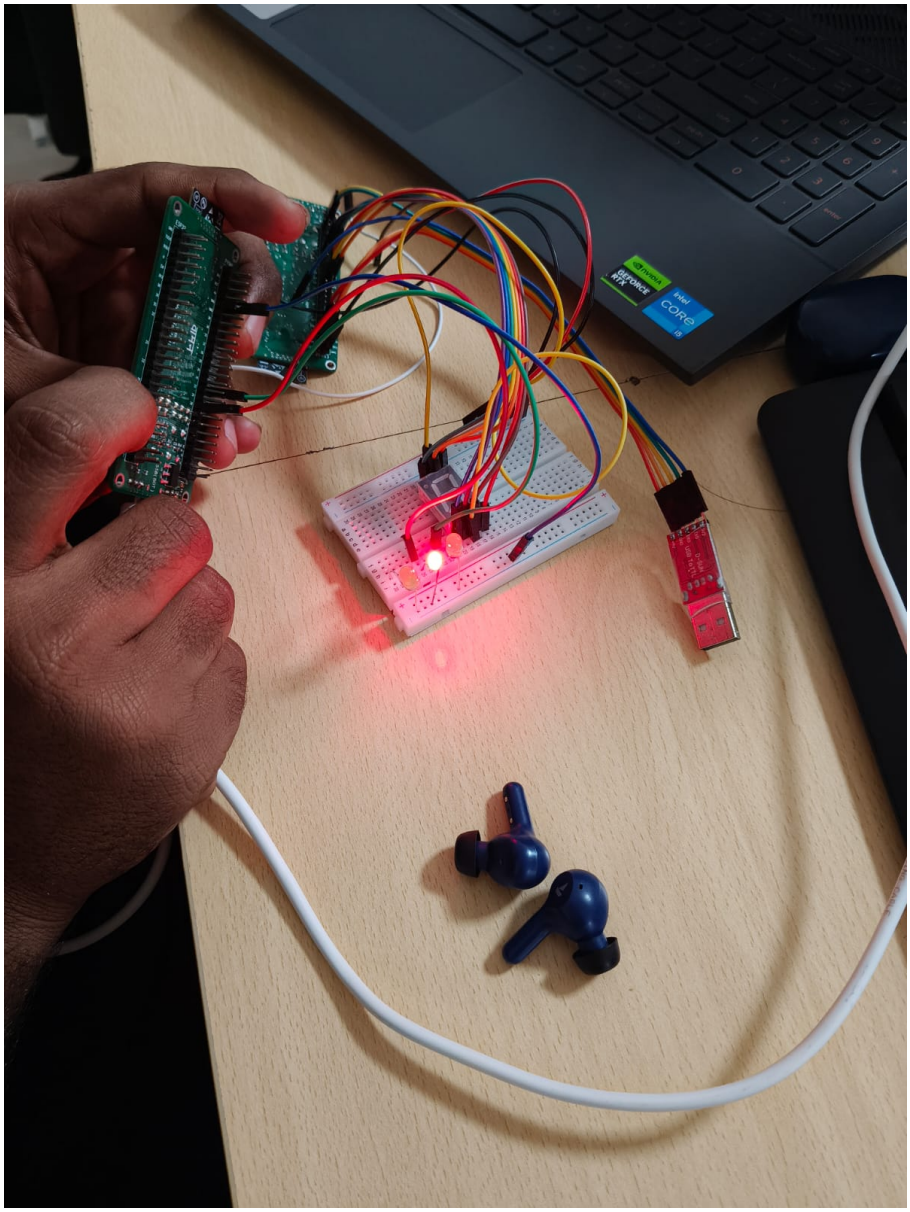
- Demonstrates low-level hardware programming.
- Eliminates dependency on external libraries.
- Provides better understanding of ARM peripheral registers.
- Faster execution due to direct register access.
- Useful for embedded systems and firmware development.

14 Applications

- Embedded System Development
- Device Driver Development
- GPIO Interface Testing
- ARM Firmware Development
- Hardware Validation
- Industrial Automation

15 Result

The experiment was successfully implemented on the VAMAN ARM Development Board using direct GPIO register programming. The GPIO Direction Register and GPIO Output Register were configured correctly, allowing three LEDs connected to GPIO18, GPIO21 and GPIO22 to blink sequentially. The experiment verified the successful operation of memory-mapped GPIO control and demonstrated the fundamentals of low-level ARM embedded programming.


GeForce
RTX
intel
CORE
i5